AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

A zoo manages various animals within different exhibits and keeps records of their health and feeding schedules. Each animal has animal ID, species, age, and health status. Each exhibit has exhibit ID, type, and capacity. Each keeper has keeper ID, name, and experience level.

a) Identify the main entities and their attributes. *(17)*

b) Draw the ER/EER diagram showing relationships and cardinalities. *(3)*

Question Q2 (15 marks)

Consider the following relation schema: Flight(FlightID, Airline, DepartureCity, ArrivalCity, DepartureTime, ArrivalTime). The functional dependencies are given by F = {FlightID -> Airline, DepartureCity, ArrivalCity, DepartureTime, ArrivalTime; Airline, DepartureCity -> ArrivalCity, DepartureTime, ArrivalTime}. Analyze this relation and its dependencies for normalization purposes.

a) Draw the functional dependency diagram for the relation Flight. *(1)*

b) Use the attribute closure and identify the primary key for the relation Flight. *(1)*

c) If we insert a row <FL123, Delta, Atlanta, New York, 10:00, 12:00> into the relation, what is the update anomaly that is violated? Give reasons. *(2)*

d) Assume that the relation has a row <FL456, Southwest, Miami, Los Angeles, 15:00, 17:30>. If we remove the flight FL456, what is the possible update anomaly that can be violated? Give reasons. *(2)*

e) Using the above functional dependencies, design a set of 3NF relations for the relation Flight. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. *(3)*

f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. *(2)*

g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. *(2)*

h) Are the 3NF relations derived in part (e) in BCNF? If not, bring the relations to BCNF. *(2)*

Question Q3 (25 marks)

Given a restaurant database with tables: Customers (id, name, email), Orders (id, customer_id, date), Products (id, name, price).

a) Write a SQL query to retrieve customer orders with order details. *(2)*

b) Create a SQL query with appropriate joins between multiple tables. *(2)*

c) Write a SQL query with aggregation functions (COUNT, SUM, AVG). *(2)*

d) Design a SQL query with subqueries for complex data retrieval. *(3)*

e) Explain the SQL query execution plan and optimization strategies. *(2)*

f) Write a SQL query with GROUP BY and HAVING clauses. *(3)*

g) Create a SQL query with window functions for analytical operations. *(5)*

h) Write a SQL query to filter records based on specific conditions. *(3)*

i) Create a SQL query to sort and limit results. *(3)*

Question Q4 (40 marks)

A zoo manages various animals within different exhibits and keeps records of their health and feeding schedules. Each animal has animal ID, species, age, and health status. Each exhibit has exhibit ID, type, and capacity. Each keeper has keeper ID, name, and experience level.

a) Identify the main entities and their attributes. *(8)*

b) Explain the relationships between entities. *(8)*

c) Describe the key concepts related to GENERAL_THEORY. *(8)*

d) Analyze the key concepts related to GENERAL_THEORY. *(8)*

e) Design the key concepts related to GENERAL_THEORY. *(8)*